Major Project Report

on

Robo-Root

Submitted by

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In partial fulfilment of the requirements for the award of degree of Bachelor of Technology in Computer Science and Engineering.



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OCTOBER 2026

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CERTIFICATE

Certified that this is a Major Project Report titled

Robo-Root

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DECLARATION

We, Mihal Palakkal, Rose Maria, Ruvais P, Mohamed Jaseeb P J, hereby declare that the project titled “Robo-Root” is a bona fide work carried out by us during the year 2026, under the guidance of Ms.Amrutha S Nair,Department of Computer Science and Engineering, SOE, CUSAT. We further declare that this project has not been submitted, either in part or full, for the award of any degree, diploma, associateship, fellowship, or any other academic recognition in any university or institution.

ACKNOWLEDGEMENT

We would like to express our sincere gratitude to everyone who supported and guided us throughout the initial phase of this project. We express our heartfelt thanks to our project mentor, Ms. Amrutha S Nair, for her valuable guidance, encouragement, and suggestions throughout the planning and development of this work. Her support helped us understand the project requirements and organise our work effectively. We are grateful to Mr. V Damodaran, Head of Division, for providing the academic support and facilities necessary for carrying out this project. We also express our sincere gratitude to Dr. Sheena S, our Project Coordinator, for her valuable guidance, support, and encouragement throughout the project. We also thank our teachers, classmates, and everyone who contributed ideas, feedback, and encouragement during the preparation of this report. Finally, we express our gratitude to our families for their constant support, patience, and encouragement throughout our academic journey.

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ABSTRACT

Plant-root-inspired tip-growing robots offer a promising approach for accessing confined environments by extending their bodies through tip eversion rather than pushing an entire rigid structure forward. This project focuses on the design and development of a pneumatic tip-growing robot that integrates controlled growth, tendon-based steering, and camera-based object detection within a modular robotic system. The primary objective is to develop a laboratory prototype capable of controlled deployment and directional movement, with particular consideration given to operation in loose, dry, noncohesive granular material.

The proposed mechanical system uses pneumatic pressure to drive the eversion of a flexible tubular body. Four steering tendons arranged around the robot provide directional control, actuated by four servo motors mounted at the base station. A camera positioned near the growing tip captures images of the surrounding environment. These images are processed using a pretrained You Only Look Once (YOLO) object detection model to identify recognizable objects and potential obstacles. OpenCV can support image acquisition and preprocessing, while the Robot Operating System 2 (ROS 2) framework provides the software architecture for coordinating perception, navigation logic, and hardware-control components.

The system is organized into modular subsystems for mechanical growth, pneumatic actuation, tendon steering, visual perception, electronic control, and software integration. This modular approach supports the independent development and integration of the robot's major functions. The design is informed by a review of existing research on soft growing robots, including pneumatic eversion, steering mechanisms, retrieval techniques, working-channel design, sensing, and control. The literature indicates that reliable operation depends on the interaction between growth, steering, payload integration, and recovery, particularly when the robot operates in a constrained environment.

The project is intended to establish an integrated laboratory prototype and provide a basis for evaluating its deployment, steering, object-detection performance, and retrieval reliability under defined test conditions. Experimental assessment should measure growth distance, deployment time, steering accuracy, complete deployment-and-retrieval success, and relevant pneumatic operating parameters. As experimental results have not yet been established, the proposed system's performance and suitability for autonomous navigation remain to be validated. The project aims to demonstrate a practical integration of pneumatic tip growth, tendon-based steering, and vision-assisted robotic control, providing a foundation for future development of soft robots for confined-space access and granular-environment exploration.

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CHAPTER 1

INTRODUCTION

Robotics has become an important area of research and development due to its applications in exploration, inspection, navigation, and operation in environments that are difficult for humans or conventional robotic systems to access. Traditional mobile robots generally rely on wheels, tracks, or articulated legs to move through their surroundings. Although these mechanisms are effective in many applications, their movement can be restricted in narrow passages, confined spaces, and environments with complex geometries. These limitations have encouraged research into flexible robotic systems capable of adapting their shape and movement to the surrounding environment.

One promising approach is the development of tip-growing robots. Unlike conventional robots that move their entire body forward, a tip-growing robot advances by extending a flexible body through a process known as eversion. In a pneumatically actuated tip-growing robot, pressurized air drives the eversion of a flexible tube, allowing new sections of the body to extend from the growing tip. This approach can reduce sliding friction along portions of the robot body and may enable navigation through narrow or restricted spaces.

However, forward growth alone does not provide sufficient directional control. A growing robot must also be capable of steering its tip toward a desired direction. To address this requirement, the proposed project incorporates four axial steering tendons arranged around the flexible tube. These tendons are actuated by four servo motors mounted at the base station. By controlling the displacement and tension of the tendons, the robot can direct the growing tip toward different directions in the horizontal X–Y plane.

In addition to mechanical steering, visual perception can improve the robot's ability to interpret its surroundings. A camera mounted near the growing tip can capture images of the environment, while a pretrained You Only Look Once (YOLO) object detection model can identify objects belonging to its supported classes. The detected objects can provide visual information for navigation and potential obstacle-aware path planning.

The Robot Operating System 2 (ROS 2) provides the decentralized middleware and computational graph framework for integrating camera processing, object detection, path planning, steering control, and pneumatic growth control. By leveraging the Data Distribution Service (DDS) communication standard, ROS 2 enables modular publish–subscribe node interactions, standardized message interfaces, and deterministic multi-threaded execution, combining pneumatic actuation, tendon-based steering, and computer vision into a cohesive robotic architecture.

1.1 Motivation for the Work

The primary motivation for this project is to investigate a robotic system that combines continuous growth with controlled directional movement and camera-assisted perception. A flexible growing robot may be useful in applications where conventional rigid robots encounter difficulties moving through narrow or constrained environments.

Pneumatic actuation provides a mechanism for extending the flexible body, while the four-tendon steering arrangement provides a method for controlling the direction of the growing tip. Locating the servo motors at the base station reduces the mass of the actuation hardware carried by the moving tip and provides convenient access to the motors and their control electronics. However, tendon routing, friction, and coordinated cable payout must be considered to ensure that the steering mechanism does not interfere with continued growth.

Another motivation is the integration of visual perception with robotic control. The tip-mounted camera provides images of the region ahead of the robot. A pretrained YOLO model can process these images to detect recognizable objects and generate bounding boxes, class labels, and confidence scores. This information can then be supplied to a ROS 2 navigation component to support decisions about the robot's direction of movement.

To unify these heterogeneous mechanical, electronic, and computational subsystems, the Robot Operating System 2 (ROS 2) was adopted as the core software integration framework. Entering into this project, ROS 2 was a new and advanced robotics technology with which the team actively familiarised itself. The framework replaces disjointed, monolithic scripts with an industry-standard, modular architecture where image processing, path planning, servo control, and pneumatic actuation execute as independent, concurrent nodes. This modularity not only ensures deterministic real-time communication and fault isolation, but also provides an extensible foundation for future advancements such as dynamic trajectory replanning and closed-loop feedback steering.

1.2 Problem Statement

The main problem addressed in this project is the design of a pneumatically actuated tip-growing robot that can extend its flexible body while controlling the direction of its growing tip. A conventional pneumatic growing tube can extend forward, but without an appropriate steering mechanism, its direction may be difficult to regulate.

The project addresses this limitation by introducing four axial steering tendons controlled by base-mounted servo motors. The system also incorporates a tip-mounted camera and a pretrained YOLO model to identify visible objects in the surrounding environment. ROS 2 is used to connect the perception and navigation components with the steering and pneumatic control systems.

The overall challenge is to coordinate continuous pneumatic growth, tendon movement, visual perception, and path planning while maintaining a mechanically feasible and controllable robot structure. The proposed design and software architecture establish a foundation for investigating these functions through simulation and experimental testing.

1.3 Objectives of the Project

The main objective is to develop a conceptual and implementation framework for a pneumatically actuated, tip-growing robot with tendon-based steering and camera-assisted navigation.

The specific objectives are:

1. To design a flexible pneumatic tip-everting mechanism capable of forward growth.
2. To develop a four-tendon steering arrangement for controlling the growing tip in the X–Y plane.
3. To position four servo motors at the base station for tendon actuation and controlled cable payout.
4. To integrate a camera near the growing tip for capturing images of the surrounding environment.
5. To use a pretrained YOLO object detection model to identify objects supported by the selected model.
6. To develop a ROS 2 software architecture connecting perception, path planning, steering control, and pneumatic growth control.
7. To develop a simulation workflow for representing planned trajectories and evaluating steering commands.
8. To establish a basis for future experimental evaluation of growth performance, steering accuracy, and visual detection reliability.

1.4 Scope of the Work

The scope of this project includes the mechanical design of the pneumatic growing tube, the four-tendon steering mechanism, the base-mounted servo arrangement, and the pneumatic control system. The electronic control architecture includes a suitable microcontroller or hardware interface for servo actuation, pneumatic valve control, and sensor communication.

The perception component focuses on camera image acquisition and object detection using a pretrained YOLO model. The software component uses ROS 2 to organize the image-processing pipeline, path-planning logic, tendon-steering commands, and pneumatic growth commands into separate functional modules.

The proposed system also considers simulation of robot trajectories and the integration of visual information into navigation decisions. The use of a pretrained model avoids the immediate requirement to train a new object detection model, although its suitability for the intended environment must be evaluated.

The project scope does not assume that object detection alone provides accurate three-dimensional obstacle positions. Additional depth sensing, camera calibration, or geometric estimation may be required for distance-aware navigation. Similarly, actual growth performance, steering precision, and obstacle-avoidance capability must be confirmed through implementation and testing rather than assumed from the proposed architecture.

1.5 Expected Outcomes

The expected outcome is a modular design for a pneumatic tip-growing robot that combines forward extension, four-tendon directional steering, and camera-based object detection.

The mechanical design is intended to provide a framework for controlling the growing tip using servo-driven tendons while the pneumatic system supports tube eversion. The software architecture is intended to connect camera images and YOLO detection results with ROS 2 path planning and low-level control.

A simulation environment can be used to examine planned trajectories and steering commands before physical testing. Following implementation, experiments can be conducted to evaluate the system's growth behavior, steering response, detection performance, and integration between the mechanical and software components.

These outcomes provide a basis for further development of flexible growing robots for inspection, exploration, and navigation in constrained environments.

1.6 Organization of the Report

The report is organized into eight main chapters. Chapter 1 introduces the background, motivation, problem statement, objectives, scope, and expected outcomes of the project. Chapter 2 presents the problem statement in greater detail and identifies the engineering challenges addressed by the proposed system. Chapter 3 reviews relevant literature on tip-growing robots, pneumatic actuation, tendon-based steering, computer vision, and robotic navigation.

Chapter 4 describes the image data and preprocessing pipeline associated with the pretrained YOLO model. Chapter 5 explains the overall system workflow, from camera image acquisition and object detection to path planning and actuation. Chapter 6 presents the modular structure of the mechanical, electronic, and software components. Chapter 7 lists the tools, platforms, libraries, and technologies used in the project. Chapter 8 summarizes the work, discusses its limitations, and identifies possible directions for future development. The report concludes with references to the research papers and technical documentation used.

CHAPTER 2

PROBLEM STATEMENT

Robotic systems designed for confined and complex environments require mechanisms that can move through restricted spaces while maintaining control over their direction. Conventional rigid robots may encounter limitations when navigating narrow passages or environments with changing geometries. Pneumatic tip-growing robots offer a potential alternative because they extend a flexible tube through eversion rather than moving the entire body forward.

Despite this advantage, controlling the direction of a growing robot remains a significant engineering challenge. Pneumatic pressure can support forward growth, but it does not independently provide precise directional steering. In addition, navigating an unfamiliar environment requires information about visible obstacles and a method for translating that information into suitable steering commands.

This project addresses these challenges through the proposed integration of pneumatic growth, four-tendon steering, camera-based object detection, and ROS 2-based control.

2.1 Definition of the Problem

The primary problem is to develop a control architecture for a pneumatically actuated tip-growing robot that can extend its flexible body and direct its growing tip toward a desired direction.

The robot uses four axial tendons arranged around the flexible tube. Four servo motors positioned at the base station regulate tendon movement to produce directional bending at the growing tip. At the same time, a pneumatic system supplies air to support the eversion process.

A camera mounted near the growing tip provides images of the surrounding environment. A pretrained YOLO model processes these images to detect objects belonging to its supported classes. The detection information is intended to support path planning through ROS 2.

The central engineering challenge is coordinating the mechanical, pneumatic, perception, and control components so that the robot's growth and steering commands are compatible with the intended navigation path.

2.2 Existing Challenges

2.2.1 Directional Control During Growth

A pneumatic growing tube may extend forward without providing sufficient control over its direction. A steering mechanism is therefore required to guide the growing tip toward a desired position.

2.2.2 Tendon Actuation and Routing

Four steering tendons must be coordinated to produce the intended tip movement. Tendon friction, tension, cable payout, and mechanical routing may affect the relationship between servo movement and actual tip direction. The design must also allow the robot to continue growing without obstructing the eversion mechanism.

2.2.3 Environmental Perception

The robot requires information about objects in front of its growing tip. Camera images can provide visual information, but detection reliability may be affected by lighting, occlusion, image quality, and differences between the operating environment and the data used to train the pretrained model.

2.2.4 Path Planning and Control Integration

Object detection alone does not determine a safe movement path. The detected objects must be interpreted by a path-planning component, which then generates a desired direction for the steering controller. The steering controller must translate that direction into suitable tendon commands while the pneumatic controller manages growth.

2.2.5 Software and Hardware Coordination

The camera, object detection model, path planner, servo motors, and pneumatic components operate as distinct functional elements. A modular ROS 2 architecture is needed to organize communication between these elements and simplify integration, testing, and debugging.

2.3 Proposed Solution

The proposed solution is a modular pneumatic tip-growing robot that combines forward extension, tendon-based steering, and visual perception.

The pneumatic subsystem supplies controlled air pressure to drive tube eversion. Four servo motors mounted at the base station actuate four axial tendons to steer the growing tip in the horizontal X–Y plane. A camera near the tip captures images of the surrounding environment, and a pretrained YOLO model detects supported object classes.

ROS 2 connects the perception and control components. The object detection output is supplied to the path planner, which determines a desired direction based on the navigation strategy. The steering controller converts the desired direction into four tendon-actuation commands, while the growth controller manages the pneumatic air supply and valve.

The proposed architecture separates perception, path planning, tendon steering, and pneumatic growth control into distinct modules. This structure supports independent development and later integration with the physical robot.

2.4 Project Objectives

The problem is addressed through the following objectives:

1. Design a pneumatic tip-everting mechanism for forward growth.
2. Develop a four-tendon steering mechanism using base-mounted servo motors.
3. Integrate a tip-mounted camera with a pretrained YOLO object detection model.
4. Develop a ROS 2 architecture for connecting visual perception with path planning and control.
5. Convert desired steering directions into coordinated tendon commands.
6. Integrate pneumatic growth control with the steering system.
7. Develop a simulation workflow for examining planned paths and steering commands.
8. Establish an experimental framework for evaluating growth, steering, and object detection performance.

2.5 Scope and Limitations

The project focuses on the mechanical concept, control architecture, visual perception pipeline, and ROS 2 integration of the proposed robot. The use of a pretrained YOLO model provides an initial object detection capability without requiring the immediate training of a new model.

However, the selected model may not recognize every obstacle relevant to the operating environment. Furthermore, two-dimensional bounding boxes do not directly provide accurate obstacle distance or three-dimensional position. Additional sensing or geometric estimation may be necessary for reliable distance-aware navigation.

The relationship between tendon displacement and tip direction must also be established through mechanical analysis and calibration. Pneumatic pressure, membrane properties, friction, and steering response must be evaluated before the robot can be considered suitable for reliable autonomous navigation.

The project therefore establishes a design and integration framework; the actual performance of the complete system must be confirmed through simulation and physical experiments.

CHAPTER 3

LITERATURE REVIEW

Plant-root-inspired growing robots have attracted research interest because of their ability to extend through confined spaces using flexible bodies. Unlike conventional mobile robots, tip-growing robots advance by everting or extending a flexible structure at the growing tip. Their potential applications include inspection, exploration, and operation in environments where conventional rigid robotic systems may face movement constraints.

Several approaches have been investigated to improve the growth, steering, sensing, and retrieval capabilities of these robots. These include pneumatic eversion, flexible structural backbones, internal working channels, magnetic actuation, shape sensing, physics-based simulation, and learning-based locomotion.

This chapter reviews ten IEEE publications relevant to soft growing robots and related burrowing systems. The studies are organized according to their principal contributions, experimental findings, and limitations. Their findings provide a foundation for identifying suitable mechanical and control approaches for the proposed pneumatic tip-growing robot with four-tendon steering, camera-based perception, and ROS 2 integration.

3.1 Review of Existing Literature

3.1.1 Large-Scale Vine Robots for Industrial Inspection

Heap et al. [1] investigated large-scale vine robots for industrial inspection using tip-growing and retractable structures. Their robot demonstrated growth and retraction over substantial distances in laboratory experiments. The study also examined traversal through an industrial pipe route and the effect of carrying a sensor mount.

The results showed that the robot could traverse a 24 m pipe route without its sensor mount. With the mount installed, it delivered video through the horizontal section but stalled at an elbow. Pressure-sensor drift also limited validation of the complete model.

This study demonstrates the potential of tip-growing robots for inspection and highlights the importance of testing the complete system with its intended payload. For the proposed project, this finding supports evaluating the camera, wiring, and other tip-mounted components as part of the integrated robot rather than testing only the empty flexible tube.

3.1.2 Material Scrunching for Internal Working Channels

Girerd et al. [2] introduced a material-scrunching approach to create working channels within miniaturized vine-inspired robots. The approach reduces friction associated with material moving through the internal channel and enables continuous access toward the growing tip.

Their reported prototype achieved 1,500 mm of growth at 55 kPa, compared with 36 mm for the standard design under the same pressure conditions. The study demonstrates that the internal arrangement of the flexible material can significantly influence growth performance.

However, the study also identified limitations related to the geometry of the tip, working-channel clearance, and the need to push the channel during operation. Retraction and more repeatable scrunching remained areas for further development.

This work highlights the importance of minimizing friction and maintaining sufficient internal clearance. These considerations are relevant when designing tendon routing, sensor wiring, and pneumatic connections for a growing robot.

3.1.3 Retractable Soft Growing Robot with a Flexible Backbone

Pi et al. [3] investigated a retractable soft growing robot incorporating a flexible backbone to provide axial support during retrieval. The backbone was intended to reduce buckling as the robot was pulled back through its deployed path.

The study reported that all 80 attempts in the straight-length experiment avoided buckling when the backbone was used. However, retrieval through curved paths still produced base crumpling and path deviation due to friction. The backbone also introduced limitations in reach and housing size.

The findings demonstrate that a mechanism that performs well along a straight path may behave differently when the robot encounters bends. For the proposed system, this suggests that steering performance should be evaluated under different path geometries and that the mechanical design should account for friction and deformation during both deployment and retrieval.

3.1.4 Pneumatic Self-Retraction of Soft Growing Robots

Kim et al. [4] proposed a self-retraction mechanism using a pressurized retraction channel and sealing ring while depressurizing the main robot body. The approach was designed to improve retrieval compared with conventional tail-pulling methods.

In the reported tests, the mechanism achieved successful retraction in all ten trials at each tested angle up to 90°, and in seven of ten trials at 120°. Conventional tail pulling performed less effectively at the tested bends. The study also demonstrated the integration of wires, a tip mount, and steering.

Despite these improvements, the mechanism remained susceptible to wrinkles, channel blockage, and restrictions caused by narrow gaps. The results emphasize that reliable retrieval requires careful consideration of channel geometry and sealing arrangements.

For the proposed pneumatic robot, this work provides useful background on the interaction between the main growth chamber, internal channels, and additional components. It also shows why deployment and retrieval should be treated as separate performance requirements.

3.1.5 Inflatable Structures for Securing Working Channels

Seo et al. [5] presented an inflatable-structure-based mechanism for maintaining a working channel within a soft growing robot. The design used multiple sub-vines to inflate and advance a sleeve around an atmospheric-pressure channel.

The researchers demonstrated tool transfer, steering, and growth. However, the measured channel diameter was smaller than its designed value, and the tests showed pressure-dependent friction in larger pipes. The presence of stiff tools also affected steering, while the additional structural components increased system complexity.

This research highlights a design trade-off between providing sufficient space for sensors or tools and maintaining the flexibility required for steering. For the proposed project, the camera wiring, tendon routing, and pneumatic pathway should therefore be arranged so that they do not unnecessarily increase the stiffness or minimum passage size of the robot.

3.1.6 Magnetic Soft Growing Robot with Shape Sensing

Calmé et al. [6] developed a magnetoelastic soft growing robot that used magnetic actuation, a camera, Hall-effect sensors, and a sampling brush. The system demonstrated controlled growth and shape reconstruction at a small scale.

A sensor stream sampled at 500 Hz supported planar shape reconstruction with a reported root-mean-square error of 2.7 mm. The study also demonstrated an ex vivo sampling application. However, the mechanism required a strong external magnetic field and close proximity to the external magnet. The reported sensing performance was limited to the evaluated configuration, and retraction was not studied.

This work demonstrates the value of sensing and feedback for understanding the shape of a growing robot. Although the proposed project uses pneumatic actuation and tendon steering rather than magnetic actuation, the general importance of measuring robot motion is relevant. Camera observations and, where appropriate, an IMU may support future feedback-based steering. Their accuracy and suitability must be established experimentally.

3.1.7 Survey of Tip-Growing Robots

Al Harthy et al. [7] presented a field survey of tip-growing robots covering fabrication, growth mechanisms, steering, sensing, control, modeling, and applications. The review examined 158 relevant manuscripts and organized the major approaches used in the field.

The survey provides a broad foundation for understanding the different design choices available for growing robots. It also shows that tip growth is an established research area, with multiple approaches already developed for steering, sensing, and retrieval.

For the proposed project, this survey is useful for positioning the design within the existing field. The contribution should not be framed as the first robot capable of tip growth or steering. Instead, the project should focus on its particular integration of pneumatic growth, four-tendon steering, visual object detection, and ROS 2-based control, with any claimed benefits evaluated through implementation and testing.

3.1.8 Autonomous Burrowing and Retrieval of Soft Robotic Anchors

Li et al. [8] investigated burrowing and retrieval of soft robotic anchors in loosely packed glass beads. Their design used vibration and changes in internal volume to influence movement and anchoring performance.

The study demonstrated downward movement and upward retrieval under specific laboratory conditions. However, the experiments used tether compensation, and downward and upward movement were evaluated separately. The addition of pouches substantially slowed the downward motion in the tested configuration.

The findings illustrate the importance of considering the surrounding medium and external assistance when evaluating burrowing performance. They also show that a successful downward movement does not automatically demonstrate reliable retrieval or completion of a full operating cycle.

Although the proposed project focuses on pneumatic tip growth and tendon steering, this study provides useful context for future testing in granular environments. Such testing would require clearly defined substrate properties, repeatable preparation conditions, and separate measurements of deployment and recovery.

3.1.9 Physics-Grounded Differentiable Simulation for Soft Growing Robots

Chen et al. [9] developed a physics-grounded differentiable simulation approach using a nonlinear stiffness model that accounts for wrinkling. The model was combined with a batched simulation of a growing chain of links and evaluated using physical demonstrations in multiple environments.

The results supported parameter fitting and comparison with unseen test configurations. However, the simulation remained computationally demanding, and high stiffness could cause instability. The planar contact formulation also did not model granular interactions or air fluidization.

This research demonstrates the value of calibrating a simulation model against physical observations rather than relying solely on an idealized geometric representation. For the proposed project, simulation can support path visualization and steering development, but the behavior of the flexible body and tendon mechanism should be validated against physical measurements when available.

3.1.10 Machine Learning-Driven Burrowing with a Snake-Like Robot

Even et al. [10] investigated learning-driven locomotion using a modular snake-like robot. Their approach used experimental runs to generate data and an optimization process to produce a sequence of motor positions for reaching a target in a controlled test environment.

The reported experiments used packing peanuts and a magnet array for height estimation. The robot reached the specified target height within the reported tolerance in the evaluated trials. However, the optimization process required substantial computation, and transfer to natural soil was not established.

This study illustrates how data-driven methods can be used to develop robot movement strategies under specific experimental conditions. However, the system differs from the proposed pneumatic tip-growing robot in its locomotion mechanism and learning approach.

The proposed project instead uses a pretrained YOLO model for visual object detection. YOLO provides object classes and bounding boxes; it does not directly generate a validated steering trajectory. Additional path-planning and control logic is required to translate the detection results into robot movement commands.

3.2 Comparative Analysis of the Reviewed Papers

The ten publications address different aspects of growing and burrowing robotics. Their findings should be compared according to their intended tasks and experimental conditions rather than ranked using a single performance measure.

The studies by Heap et al. [1], Girerd et al. [2], Pi et al. [3], Kim et al. [4], and Seo et al. [5] primarily highlight the relationships among growth, internal access, steering, payload integration, and retrieval. Calmé et al. [6] and Al Harthy et al. [7] provide insights into sensing and the broader design landscape. Li et al. [8] investigate movement and anchoring in granular media, while Chen et al. [9] and Even et al. [10] examine simulation and learning-based movement strategies.

Several common engineering considerations emerge from this comparison:

1. **Mechanical integration:** Adding a camera, wiring, tools, or other payloads can affect growth and path completion.
2. **Steering and curvature:** A mechanism that performs well on a straight path may encounter friction, deformation, or retrieval problems on curved paths.
3. **Internal clearance:** Additional channels and components can improve access but may increase friction, complexity, or minimum passage size.
4. **Sensing and feedback:** Sensor measurements and model predictions should be validated against independent observations.
5. **Simulation and experimental testing:** Simulation is useful for design and planning, but its conclusions are limited by the accuracy of the model and the conditions represented.
6. **Environmental dependence:** Results obtained in pipes, glass beads, packing peanuts, or other controlled environments cannot automatically be generalized to natural soil or different operating conditions.

These findings suggest that the mechanical design, sensing arrangement, and control system should be evaluated as an integrated system rather than as isolated components.

3.3 Research Gaps and Motivation for the Proposed Project

The reviewed literature demonstrates that tip-growing mechanisms, tendon or other steering approaches, sensing, and retrieval strategies have already been investigated. Therefore, the research gap for the proposed project should be defined around the particular system architecture and its evaluation, rather than claiming that these general capabilities are absent from previous work.

The first gap concerns the integration of pneumatic growth with four-tendon steering. The proposed design requires coordinated tendon movement while the flexible tube continues to evert. The relationship between servo displacement, tendon tension, and actual tip direction must be established through mechanical design and calibration.

The second gap concerns visual perception and navigation. A tip-mounted camera and pretrained YOLO model can identify objects belonging to the model's supported classes. However, the detection results must be connected to a path planner and steering controller before they can influence the robot's movement. Bounding boxes alone do not establish obstacle distance or guarantee a collision-free trajectory.

The third gap concerns integrated system validation. The reviewed studies demonstrate capabilities under different mechanical arrangements and experimental conditions, but their numerical results are not directly comparable. The proposed project therefore needs its own evaluation procedure to measure steering response, detection performance, and the behavior of the integrated control architecture.

These gaps motivate the development of a modular ROS 2 architecture connecting camera acquisition, object detection, path planning, four-tendon steering, and pneumatic growth control. The proposed contribution is the design and evaluation of this particular integration, subject to the capabilities demonstrated by the completed prototype.

3.4 Summary

This chapter reviewed ten IEEE publications covering large-scale vine robots, internal working channels, retrieval mechanisms, magnetic shape sensing, the broader tip-growing robot field, granular burrowing, physics-based simulation, and learning-driven locomotion.

The literature indicates that growth, steering, payload integration, sensing, and retrieval are closely related engineering challenges. It also shows that performance depends on the robot's geometry, operating pressure, path curvature, surrounding medium, and sensing arrangement.

Based on these findings, the proposed project combines pneumatic tip growth with four-tendon steering, base-mounted servo motors, camera-based object detection using a pretrained YOLO model, and ROS 2-based control. The objective is to develop a modular system architecture and evaluate its behavior through appropriate simulation and experimental testing. The project should report only those performance claims that are supported by its own implementation and results.

CHAPTER 4

DATASET DESCRIPTION AND PREPROCESSING

Computer vision is an important component of the proposed pneumatic tip-growing robot because it provides visual information about the environment ahead of the growing tip. The camera captures images that are processed by a pretrained You Only Look Once (YOLO) object detection model. The detection results can be used by the navigation system to identify visible objects and support path-planning decisions.

Unlike a project that trains an object detection model from scratch, the proposed system initially uses a pretrained YOLO model. Therefore, the main focus of this chapter is image acquisition, preprocessing, model inference, and integration of detection results with the Robot Operating System 2 (ROS 2) architecture.

4.1 Dataset Description

4.1.1 Source of Image Data

The primary input to the object detection system consists of images captured by a camera mounted near the growing tip of the robot. These images represent the visible surroundings and provide information about objects that may influence the robot's movement.

During development, prerecorded images or publicly available sample images may be used to test the detection pipeline. During physical operation, images are obtained from the robot's camera through a ROS 2 image stream.

The final image source and the number of images used for testing should be documented after the implementation is established. No custom dataset size or experimental image count is assumed in the current project description.

4.1.2 Pretrained YOLO Model

YOLO is a real-time object detection method that identifies objects in an image by predicting their classes, bounding boxes, and confidence scores.

The proposed system uses pretrained model weights, allowing object detection to be implemented without initially training a new model. The model variant and its supported object classes must be selected according to the computational resources available and the objects relevant to the intended environment.

The pretrained model may identify commonly represented object categories, but it cannot be assumed to recognize every obstacle encountered by the robot. If the required objects are not supported by the selected model, additional data collection and model fine-tuning may be required.

4.1.3 Dataset Organization

Since the initial implementation uses a pretrained model, a training, validation, and testing split is not automatically required for model training. However, a separate set of representative images should be retained for evaluating detection performance.

The evaluation images should represent the expected operating conditions, including different object positions, viewing angles, lighting conditions, and possible occlusions. Images captured from the same video sequence should be handled carefully when constructing evaluation sets to avoid overly optimistic results from nearly identical frames.

4.2 Image Preprocessing

Image preprocessing converts camera images into a format suitable for the selected YOLO inference interface. The exact operations depend on the model implementation and its software library.

4.2.1 Image Acquisition

Images are captured by the camera positioned near the growing tip. The camera stream is received by the ROS 2 perception node for processing.

Image quality can be affected by illumination, focus, camera orientation, motion blur, and obstructions near the camera. These factors should be considered when positioning the camera and evaluating the detection system.

4.2.2 Image Format Conversion

The ROS 2 image message is converted into an image representation compatible with the inference library. Depending on the camera driver and software interface, this may involve converting the image message to an OpenCV image and checking the color-channel order.

Correct image encoding is important because an incorrect color format can affect the model input and the interpretation of detection results.

4.2.3 Image Resizing and Letterboxing

The input image is resized or letterboxed to match the input dimensions required by the selected model. Letterboxing preserves the image aspect ratio by adding padding when necessary, reducing distortion of object shapes.

The actual input resolution depends on the selected YOLO model and inference configuration. A resolution such as 640×640 pixels may be used as an example if supported by the selected implementation, but it should not be reported as a finalized project setting until confirmed.

4.2.4 Pixel Normalization and Tensor Conversion

The preprocessed image is converted into the numerical representation expected by the model. Depending on the inference library, this may involve scaling pixel values, arranging image dimensions, and converting the image into a tensor.

When a standard YOLO inference interface is used, resizing, normalization, tensor conversion, and related preprocessing operations may be performed internally. Additional manual preprocessing should be avoided when the library already handles these operations.

4.2.5 Object Detection Inference

After preprocessing, the image is supplied to the pretrained YOLO model. The model generates object detection predictions, including class labels, confidence scores, and bounding-box coordinates.

A confidence threshold can be configured to filter detections, and the inference pipeline may apply non-maximum suppression to remove overlapping duplicate detections. These settings should be selected according to the chosen model and evaluated during testing.

4.2.6 Postprocessing

The model's detection output is converted into a format suitable for the ROS 2 application. The processed results may include the detected object's class, confidence score, and bounding-box coordinates relative to the image.

The detection results can then be published or forwarded to the path-planning component. Because image bounding boxes provide two-dimensional information, additional depth sensing, camera calibration, or geometric estimation may be required to calculate metric obstacle positions.

4.3 Image Processing Workflow

The proposed image processing pipeline consists of the following stages:

1. **Camera acquisition:** Capture an image of the environment near the growing tip.
2. **Format conversion:** Convert the ROS 2 image message into the representation required by the inference library.
3. **Resizing and letterboxing:** Adjust the image to the model's input dimensions while preserving its aspect ratio.
4. **Normalization and tensor conversion:** Prepare the numerical image representation for inference.
5. **YOLO inference:** Run the pretrained object detection model.

6. **Postprocessing:** Extract and filter object classes, confidence scores, and bounding boxes.
7. **ROS 2 integration:** Forward the detection results to the path-planning component.

Figure 4.1 illustrates the image processing workflow of the proposed system.

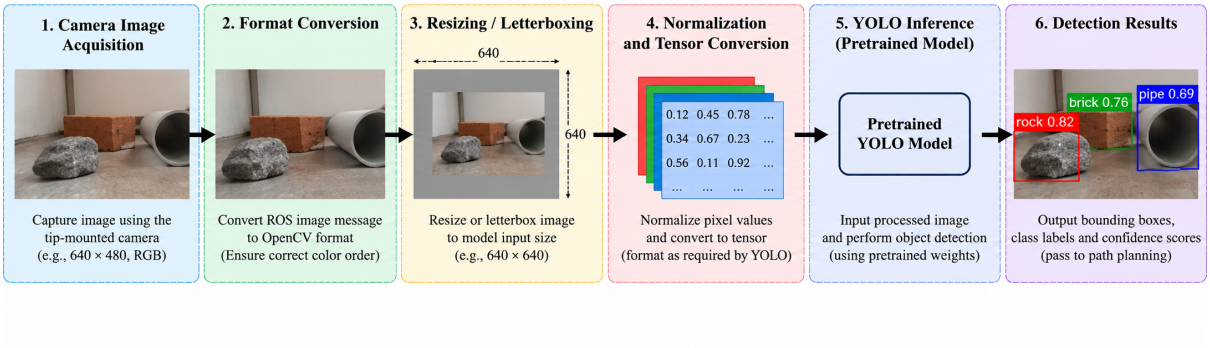


Figure 4.1: Image Processing Pipeline

This workflow separates image processing from steering and pneumatic control, allowing the perception component to be developed and evaluated independently.

4.4 Integration with ROS 2

The visual perception pipeline is integrated into the robotic system through standard ROS 2 messaging interfaces. The camera driver node acquires live frames from the tip camera and publishes them onto the `/camera/image_raw` topic using the standard `sensor_msgs/msg/Image` message structure.

The object detection node subscribes to `/camera/image_raw`. To interface the incoming ROS image messages with the computer vision libraries, the node employs the `cv_bridge` package. This utility converts the serialized ROS image payload into an OpenCV BGR NumPy matrix without introducing unnecessary copying overhead. Once the image is processed and YOLO inference is executed, the resulting object bounding boxes, class labels, and confidence probabilities are formatted into standard `vision_msgs/msg/Detection2DArray` messages and broadcast onto `/detections/bounding_boxes`.

The path-planning node subscribes to these detection arrays to extract obstacle boundaries and estimate clearance corridors relative to the camera frame. The planner derives target steering angles, which the tendon kinematics controller converts into servo commands for the four steering tendons. Simultaneously, the pneumatic growth controller regulates the air supply and valve duty cycle. This standardized ROS 2 publish–subscribe architecture completely decouples image processing from actuation, ensuring that computer vision algorithms can be modified, re-trained, or replaced without altering the low-level steering or growth controllers.

4.5 Evaluation Considerations

The pretrained model should be evaluated using images representative of the intended operating environment. Relevant measures may include precision, recall, mean Average Precision (mAP), inference time, and the frequency of missed or false detections.

For real-time operation, inference latency and processing rate are also important because delayed detections may affect navigation decisions. If the robot operates in confined or partially occluded environments, testing should specifically examine whether the selected model recognizes the relevant obstacles under those conditions.

Actual dataset statistics and performance metrics must be obtained through implementation and testing. They are not assumed in this chapter.

4.6 Limitations

The proposed image-processing system depends on the quality of the camera images and the object categories supported by the pretrained YOLO model. Changes in illumination, viewpoint, occlusion, and environmental appearance may reduce detection reliability.

Furthermore, object detection does not independently provide a complete navigation solution. The detected objects must be interpreted by the path planner, and reliable obstacle-distance estimation may require additional sensing or geometric processing.

These limitations should be considered when integrating the vision system with the pneumatic growing mechanism and four-tendon steering controller.

CHAPTER 5

SYSTEM WORKFLOW

The proposed pneumatic tip-growing robot integrates a flexible eversion mechanism, four-tendon steering, pneumatic actuation, camera-based object detection, and ROS 2-based control. The system workflow describes how these components interact to support forward growth and directional steering.

The overall system is divided into mechanical, pneumatic, perception, planning, and control functions. The pneumatic subsystem supports the extension of the flexible tube, while four servo motors located at the base station control the steering tendons. A camera positioned near the growing tip captures images of the surrounding environment, and a pretrained YOLO model processes the images to detect objects belonging to its supported classes.

ROS 2 provides the communication framework connecting the perception and control components. The resulting architecture allows each subsystem to be developed and tested independently before integration.

5.1 Overall System Operation

The system operates through a sequence of image acquisition, object detection, path planning, tendon steering, and pneumatic growth control.

Initially, the pneumatic subsystem supplies air to the flexible tube to initiate or maintain eversion. At the same time, the camera captures images of the environment ahead of the growing tip. The object detection node processes these images using the pretrained YOLO model and produces detection results.

The path-planning component interprets the available environmental information and determines a desired movement direction according to the implemented navigation strategy. The steering controller converts this direction into coordinated commands for the four servo motors.

The pneumatic controller regulates the growth command according to the intended operating sequence and available pressure feedback. During operation, updated camera images can provide further information for subsequent steering and planning decisions.

5.2 Pneumatic Growth Mechanism

The pneumatic subsystem provides the pressure required to extend the flexible growing tube through eversion. An air pump or compressor supplies air through the pneumatic circuit, which may include a regulator, control valve, pressure sensor, and connecting tubing.

The pressure regulator helps establish the intended operating pressure, while the control valve manages the air supply according to the growth command. A pressure sensor can provide

feedback to the controller for monitoring operating conditions.

The flexible tube advances as material everts at the growing tip. The actual growth rate and stability depend on the tube material, geometry, applied pressure, friction, and interaction with the surrounding environment.

The pneumatic subsystem is therefore treated as a separate control function from the tendon-steering mechanism, although both must operate in coordination during robot movement.

5.3 Four-Tendon Steering Mechanism

Directional control is provided by four axial tendons arranged around the flexible tube. Each tendon is connected to a servo-actuation mechanism positioned at the base station.

The tendons are arranged to provide steering authority in different directions in the X–Y plane. Coordinated tendon displacement and tension produce bending of the steering region near the growing tip. The direction and magnitude of bending depend on the tendon arrangement, attachment geometry, tension, and mechanical properties of the flexible structure.

The steering controller calculates the required tendon commands from the desired direction. These commands are transmitted to the servo-control hardware, which positions the servo motors accordingly.

Because the tube continues to grow during operation, the tendon mechanism must also accommodate cable payout and maintain appropriate routing and tension. The relationship between servo movement and actual tip direction must be calibrated on the physical mechanism.

5.4 Camera-Based Object Detection

The perception subsystem uses a camera mounted near the growing tip to capture images of the surrounding environment. The camera stream is made available to a ROS 2 node for image processing.

The object detection node converts each image into the format required by the selected pretrained YOLO inference interface. The image is resized or letterboxed as necessary, and the model performs inference to identify supported object classes.

The resulting detections contain class labels, confidence scores, and bounding-box coordinates. These results provide visual information about objects in the camera's field of view.

The perception subsystem does not independently determine a safe trajectory. Instead, its output is supplied to the path planner, which interprets the detections according to the navigation strategy. If accurate obstacle distance is required, additional depth sensing, calibration, or geometric estimation may be required.

5.5 ROS 2 Path Planning and Control

ROS 2 connects the perception, planning, and actuation components through a modular, deterministic software architecture. Rather than executing as a monolithic sequential script, the control framework operates as a multi-threaded computational graph managed by a ROS 2 Multi-Threaded Executor. This decouples computationally intensive neural network inference from high-frequency, time-critical actuator control loops.

The camera node continuously captures and streams video frames at 30 Hz. Upon arrival of a new frame, the object detection node processes the image using the pretrained YOLO model, extracting 2D bounding boxes, semantic classes, and confidence scores. The path-planning node subscribes to these detection arrays and queries the `tf2` coordinate transformation buffer to relate detected obstacles to the robot's moving tip frame.

Based on the spatial distribution of detected obstacles, the planner computes an obstacle-free heading vector in terms of horizontal bending angles (θ_x, θ_y) at an update rate of 10–20 Hz. The steering controller converts these angular targets into continuous length setpoints $(\Delta l_1, \Delta l_2, \Delta l_3, \Delta l_4)$ for the four tendons via an inverse kinematics mapping. These setpoints are published to the microcontroller interface at a high-frequency control loop (50–100 Hz) to ensure smooth, jitter-free servo motion.

Simultaneously, the growth-control node manages the pneumatic subsystem by monitoring chamber pressure feedback. It regulates the solenoid valve duty cycle to maintain stable eversion while synchronizing the servo spool payout rate with forward growth speed. If a tendon becomes too tight, growth is temporarily throttled; if excess slack is detected, spools are tensioned to maintain positive contact.

A watchdog monitoring thread continuously checks heartbeat signals across all active ROS 2 nodes and the microcontroller bridge. In the event of frame loss, node failure, or unexpected pressure surges, the watchdog triggers a graceful emergency stop, immediately venting pneumatic pressure and holding tendon positions.

Figure 5.1 illustrates the overall ROS 2 software architecture and hardware interface for the proposed growing robot.

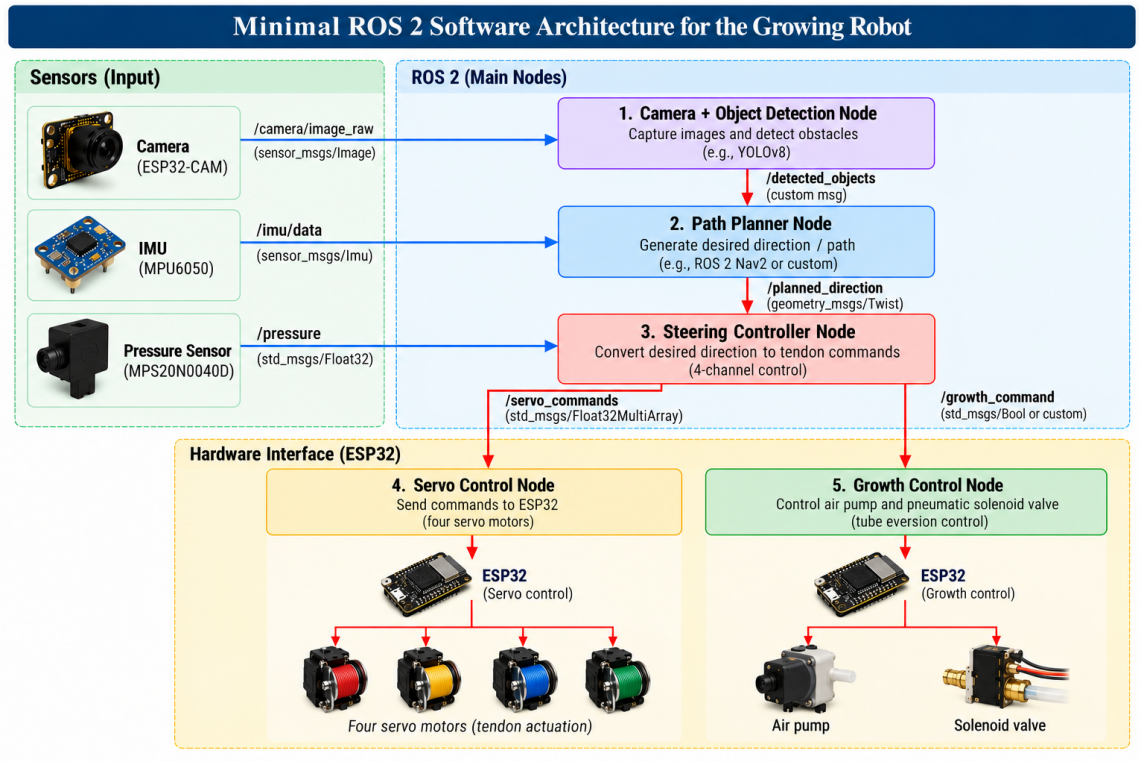


Figure 5.1: ROS 2 Software Architecture and Hardware Interface for the Growing Robot

5.6 Integrated System Workflow

The complete proposed workflow consists of the following steps:

1. **System initialization:** Initialize the camera, ROS 2 nodes, embedded controller, servo interface, and pneumatic control components.
2. **Pneumatic growth:** Supply regulated air to initiate or maintain eversion of the flexible tube.
3. **Image acquisition:** Capture images of the environment using the tip-mounted camera.
4. **Image preprocessing:** Convert the camera image into the input format required by the pretrained YOLO model.
5. **Object detection:** Run inference and obtain supported object classes, confidence scores, and bounding boxes.
6. **Path planning:** Interpret the detection results and determine a desired movement direction according to the implemented navigation strategy.
7. **Tendon steering:** Convert the desired direction into coordinated commands for the four servo motors.
8. **Growth coordination:** Manage pneumatic growth while ensuring that tendon movement and cable payout remain compatible with continued eversion.

9. **Repeated perception:** Acquire updated images and repeat the planning and steering process as required.

This workflow describes the intended system operation. The actual timing, control frequency, and feedback behavior will depend on the implemented hardware and software.

5.7 System Feedback and Safety

Feedback is important for maintaining controlled operation. A pressure sensor can monitor pneumatic pressure, while an IMU or other suitable sensing arrangement may provide information about the robot's orientation or movement. Camera observations can provide additional environmental information.

The available measurements should be used to detect conditions such as excessive pressure, stalled growth, unexpected steering response, or loss of reliable perception. Appropriate limits and shutdown behavior should be established before physical testing.

The pneumatic system should include pressure regulation and suitable overpressure protection. Servo commands and tendon tension should also be limited to avoid damaging the flexible tube or steering mechanism.

The feedback architecture should be expanded according to the sensors actually installed and validated. The presence of a sensor or software node alone does not establish that the robot has closed-loop steering or autonomous obstacle avoidance.

CHAPTER 6

MODULAR STRUCTURE

The proposed pneumatic tip-growing robot is organized into separate functional modules to simplify mechanical design, hardware integration, software development, and testing. Each module performs a specific function and communicates with the relevant modules to support the overall operation of the robot.

The system consists of a pneumatic growth module, a tendon-based steering module, a servo-actuation module, a camera-based perception module, an embedded electronics module, and a ROS 2 software module. These components work together to support forward growth, directional steering, environmental observation, and navigation planning.

A modular architecture allows the individual subsystems to be developed and evaluated independently before the complete system is integrated.

6.1 Mechanical Growth Module

The mechanical growth module consists of the flexible tube and the tip-everting structure. The flexible tube forms the growing body of the robot and extends through eversion when suitable pneumatic pressure is applied.

The geometry and material properties of the tube influence its flexibility, growth behavior, and interaction with the surrounding environment. The growing tip also provides the mounting region for the steering arrangement and camera, where the mechanical design permits.

The module must be designed so that the additional tendons, wiring, and sensor components do not prevent the tube from everting. Adequate space and suitable routing are required to reduce friction and interference during growth.

Figure 6.1 illustrates the overall mechanical structure of the pneumatic tip-growing robot, depicting the base station, the flexible tip-everting tube, and the growing tip assembly.

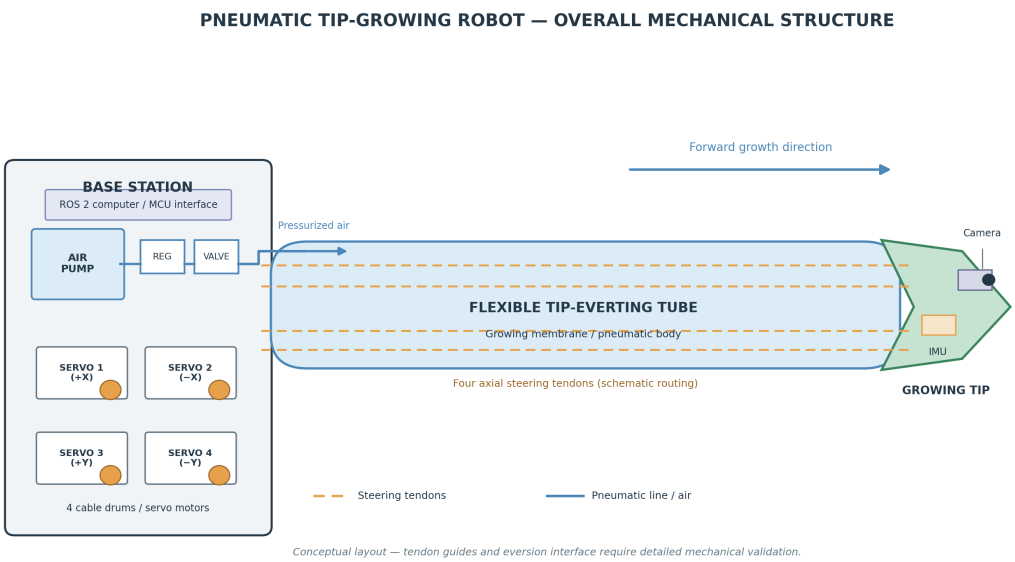


Figure 6.1: Overall Mechanical Structure of the Pneumatic Tip-Growing Robot

Main components:

- Flexible pneumatic growing tube.
- Tip-everting structure.
- Tip support and tendon attachment arrangement.
- Mechanical guides for tendon routing.

6.2 Pneumatic Actuation Module

The pneumatic module supplies and regulates the air required to drive the growing mechanism. It consists of an air pump or compressor, pneumatic tubing, pressure regulation, and a suitable control valve. A pressure sensor may be incorporated to monitor operating pressure.

The pump provides the air supply, while the regulator helps maintain the intended pressure range. The valve controls the air supply according to the growth command. Pressure measurements can be communicated to the controller for monitoring and safety functions.

The design of this module depends on the required pressure, air flow, tube geometry, and characteristics of the flexible material. Suitable pressure regulation and overpressure protection are necessary before physical testing.

Main components:

- Air pump or compressor.
- Pressure regulator.
- Pneumatic control valve.

- Pressure sensor, if implemented.
- Pneumatic hoses and fittings.

6.3 Tendon-Based Steering Module

The steering module consists of four axial tendons arranged around the flexible tube. The tendons are positioned to provide directional control of the growing tip in the horizontal X–Y plane.

The tendons are connected to the servo-actuation mechanism at the base station. Coordinated changes in tendon displacement and tension produce bending at the steering region near the growing tip.

The steering mechanism must accommodate continuous tube growth. Therefore, tendon routing, cable payout, tension control, and friction are important design considerations. The relationship between servo movement and the resulting tip direction must be determined through mechanical analysis and calibration.

Figure 6.2 illustrates the directional steering behavior of the growing robot under four-tendon control, showing the bending configurations and corresponding tendon tension levels.

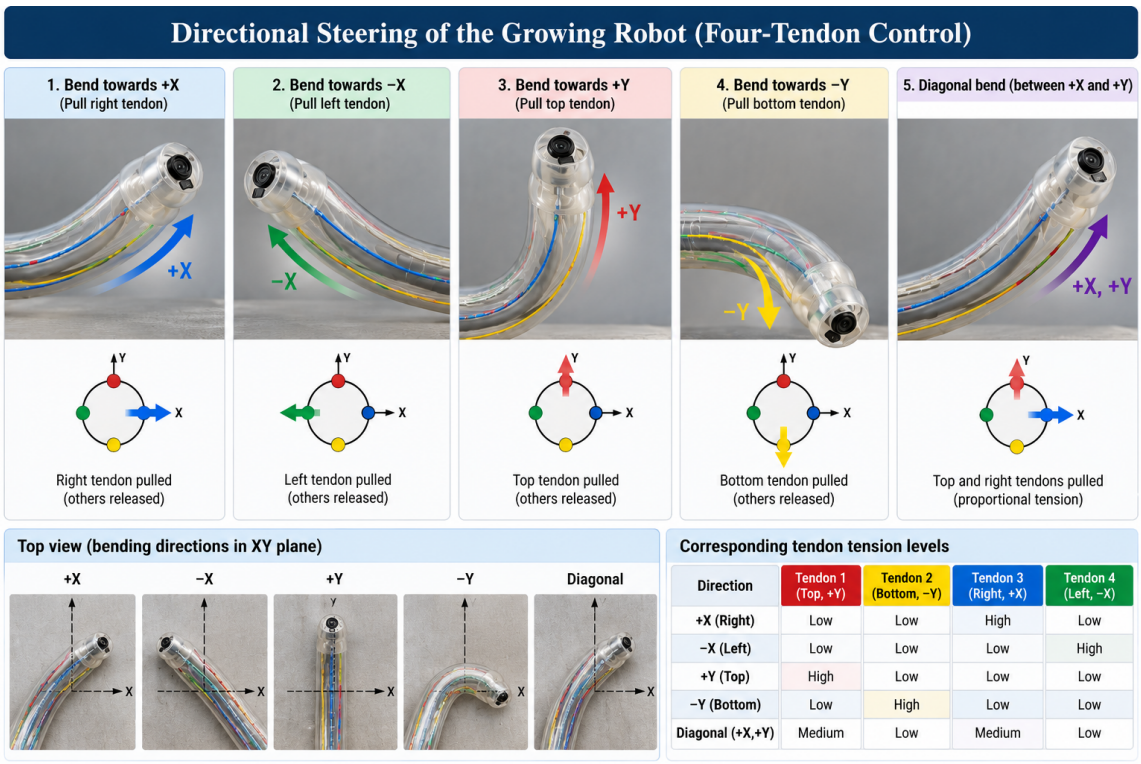


Figure 6.2: Directional Steering of the Growing Robot via Four-Tendon Control

Main components:

- Four axial steering tendons.
- Tendon guides and routing channels.

- Tendon attachment points near the steering tip.
- Base-mounted tendon actuation interfaces.

6.4 Servo Actuation Module

The servo-actuation module provides controlled movement of the four steering tendons. Four servo motors are mounted at the stationary base station rather than at the growing tip. This arrangement reduces the actuation mass carried by the moving portion of the robot.

Each servo is assigned to a tendon and receives a command from the steering controller through the embedded control interface. Servo movement adjusts tendon displacement and tension according to the desired steering direction.

The servo mounting structure must provide adequate support and alignment. The tendon drive mechanism must also prevent excessive slack, unwanted tangling, and interference with pneumatic growth.

Figure 6.3 depicts the detailed layout of the base-mounted servo motors and cable-drum assembly, including spool design, cable guides, and tendon entry into the growing tube.

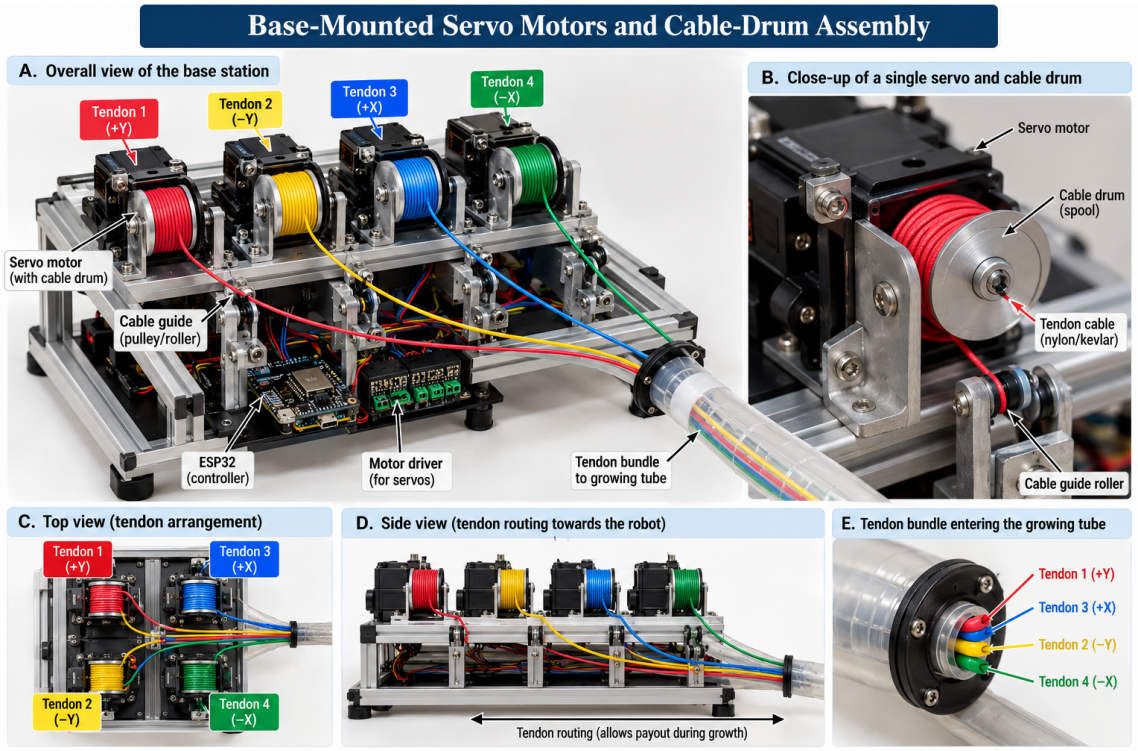


Figure 6.3: Base-Mounted Servo Motors and Cable-Drum Assembly

Main components:

- Four servo motors.
- Servo mounting frame.

- Tendon spools or cable drums, if used.
- Tendon guides and tension adjustment mechanisms.
- Suitable servo power supply.

The servo torque, angular range, spool diameter, and allowable tendon tension must be selected according to the actual mechanical requirements.

6.5 Camera-Based Perception Module

The perception module uses a camera positioned near the growing tip to capture images of the environment ahead of the robot. The images are supplied to the object detection pipeline for visual analysis.

A pretrained YOLO model processes the camera images and identifies objects belonging to its supported classes. The model produces bounding boxes, class labels, and confidence scores for the detected objects.

The detection results provide visual information for the navigation system. However, accurate obstacle distance and three-dimensional position cannot generally be obtained from bounding boxes alone. Additional sensing, calibration, or geometric estimation may be necessary for metric navigation.

The camera mount and wiring arrangement should be designed to protect the sensor and avoid interfering with tube eversion and steering.

Main components:

- Camera module.
- Camera mounting structure.
- Image acquisition interface.
- Pretrained YOLO inference software.

6.6 Embedded Electronics Module

The embedded electronics module connects the servo motors, pneumatic valve, pressure sensor, and other installed hardware to the control software. An ESP32 or another suitable microcontroller can provide low-level actuator control and sensor communication.

The embedded controller receives commands from the higher-level ROS 2 software through an appropriate communication interface. It then generates servo-control signals and operates the pneumatic valve through a correctly rated driver circuit.

If pressure sensing is implemented, the controller can also transmit pressure measurements for monitoring and control. The hardware design must account for electrical compatibility, current requirements, grounding, and actuator protection.

Figure 6.4 presents the electronics and wiring diagram of the growing robot, showing connections between the ESP32 controller, power supply, servo actuators, valve driver, and sensory devices.

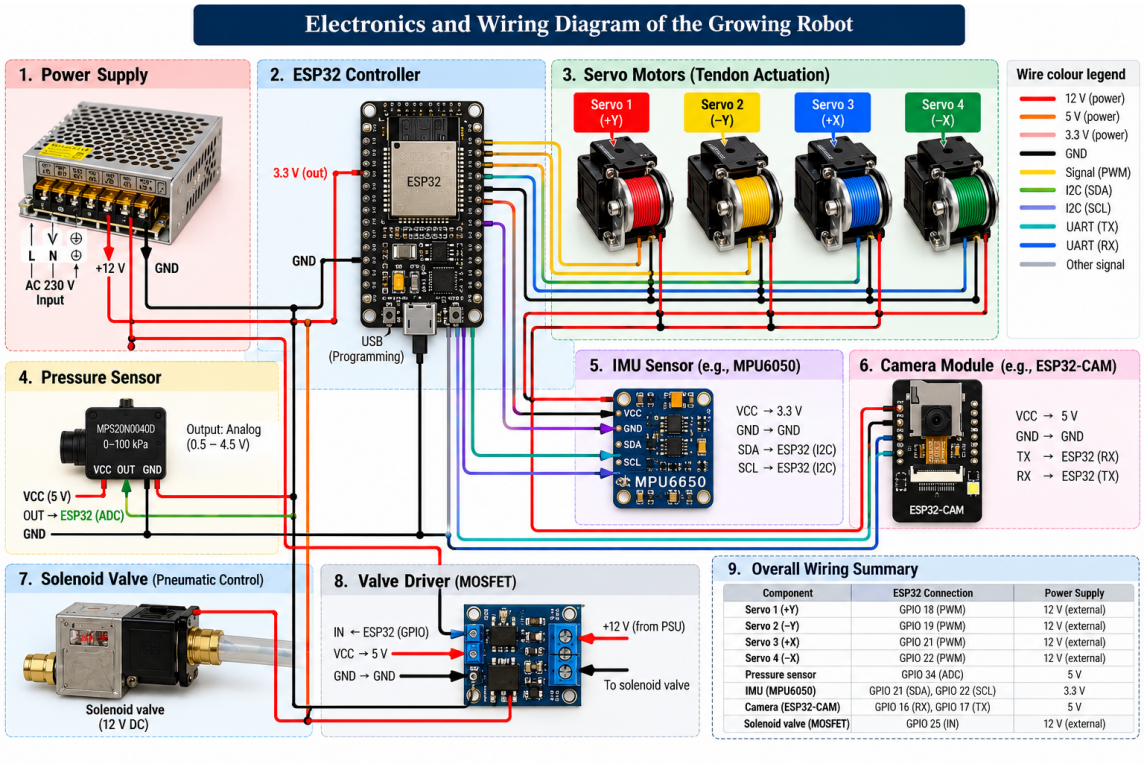


Figure 6.4: Electronics and Wiring Diagram of the Growing Robot

Main components:

- ESP32 or suitable microcontroller.
- Servo signal connections.
- External servo power supply.
- Pneumatic valve driver.
- Pressure sensor interface, if used.
- Communication interface with the ROS 2 computer.

The final wiring and pin assignments depend on the selected hardware and must be verified before assembly.

6.7 ROS 2 Software Module

The ROS 2 software module coordinates visual perception, object detection, obstacle-aware path planning, directional steering kinematics, and pneumatic growth actuation. Leveraging

the distributed publish–subscribe computational graph of ROS 2, the software is decomposed into independent, modular nodes that communicate across standardized message interfaces. This modularity decouples sensing, computation, and actuation, enabling each component to be developed, simulated, and debugged independently before complete hardware-in-the-loop deployment.

6.7.1 Node Architecture and Communication Topics

The principal functional nodes within the ROS 2 software module comprise:

1. **Camera Driver Node (`camera_driver_node`):** Interfaces directly with the tip-mounted camera hardware (via V4L2 or USB video capture). It captures visual frames at a deterministic frame rate (e.g., 30 frames per second) and publishes raw frames onto the `/camera/image_raw` topic using the standard `sensor_msgs/msg/Image` format, accompanied by camera intrinsic parameters on `/camera/camera_info`.

2. **Object Detection Node (`yolo_detector_node`):** Subscribes to `/camera/image_raw`. Using the `cv_bridge` interface library, it converts the incoming ROS image messages into OpenCV NumPy image representations, performs letterboxing and tensor normalization, and executes inference using the pretrained YOLO model. The detected objects, along with bounding-box coordinates, class labels, and confidence metrics, are published onto `/detections/bounding_boxes` using the `vision_msgs/msg/Detection2DArray` message definition.

3. **Path-Planning Node (`path_planner_node`):** Receives bounding-box detections from `/detections/bounding_boxes` and spatial transformation data from the `tf2` buffer. By evaluating the centroid and spatial span of detected obstacles within the camera's field of view, the planner calculates obstacle-free navigation vectors. It computes a target tip bending angle (θ_x, θ_y) and publishes steering commands to `/navigation/target_heading`.

4. **Tendon Kinematics and Steering Controller (`tendon_controller_node`):** Subscribes to `/navigation/target_heading`. This node executes the inverse kinematic model for the antagonistic four-tendon configuration. It maps the requested Cartesian bending angle (θ_x, θ_y) into required tendon displacements $(\Delta l_1, \Delta l_2, \Delta l_3, \Delta l_4)$ and corresponding angular setpoints for the four base-mounted servo motors. These setpoints are published onto `/actuation/servo_commands` as a multi-channel floating-point array (`std_msgs/msg/Float32MultiArray`).

5. **Pneumatic Growth Manager (`growth_manager_node`):** Manages pneumatic eversion and coordinated cable payout. It subscribes to chamber pressure measurements from `/sensors/pneumatic_pressure` (`sensor_msgs/msg/FluidPressure`) and publishes solenoid valve modulation commands onto `/actuation/valve_state`. Crucially, this node synchronizes the forward growth rate with the servo spool payout rate, preventing

cable tension buildup from stalling tube eversion or excess slack from causing cable snagging.

6. **Microcontroller Hardware Bridge (`micro_ros_bridge_node`):** Establishes the communication conduit between the host ROS 2 computation environment and the ESP32 microcontroller. Operating over Micro-ROS (Micro XRCE-DDS) via a high-speed serial UART link, this bridge maps ROS 2 topic subscriptions and publications directly to hardware pins on the microcontroller.

6.7.2 Coordinate Transformation Pipeline (`tf2`)

A specialized responsibility of the software module is tracking the dynamic morphology of the growing robot. Unlike traditional rigid link manipulators whose segment lengths are invariant, the length of the everting tube grows continuously:

- `base_link`: A fixed spatial coordinate frame anchored at the base station and spool assembly.
- `growing_body`: An intermediate coordinate frame representing the instantaneous curved centerline of the everting tube.
- `growing_tip`: The dynamic coordinate frame positioned at the leading edge of eversion, which translates forward during growth and rotates during tendon steering.
- `camera_optical_frame`: The coordinate frame of the visual sensor, rigidly attached to the growing tip.

By continuously publishing transformation updates via the `tf2` library, the software allows visual detections gathered in camera optical coordinates to be instantly expressed in base or world coordinates for consistent trajectory generation.

6.7.3 System Safety and Watchdog Monitoring

To protect delicate soft robot structures from excessive stress or pressure, the ROS 2 software module incorporates a dedicated watchdog timer. The node monitors periodic heartbeat signals exchanged between the host computer and the embedded microcontroller. If communication is interrupted, or if internal chamber pressure exceeds predefined safety thresholds, the software immediately initiates an emergency shutdown service: venting the pneumatic solenoid valve to atmospheric pressure and locking the servo spools to prevent uncontrolled runaway.

6.8 Integration of the Modules

The modules are integrated through their mechanical interfaces, electrical connections, and software communication pathways.

The pneumatic growth module extends the flexible tube. The tendon-steering module controls the direction of the growing tip through four tendons driven by the base-mounted servo motors. The perception module provides camera images and object detection results to the ROS 2 software.

The path planner determines the desired direction based on the available environmental information. The steering controller converts this direction into tendon commands, while the growth-control module manages pneumatic actuation.

The embedded controller provides the hardware interface for the servos and pneumatic valve. This separation allows the mechanical, pneumatic, electronic, and software components to be tested individually before full-system evaluation.

CHAPTER 7

TOOLS USED

The development of the proposed plant-root-inspired pneumatic tip-growing robot requires a combination of mechanical components, pneumatic actuators, electronic control hardware, computer vision, and robotics software. These tools work together to enable pneumatic growth, directional steering, environmental perception, and coordination between the robot's hardware and software components.

The tools are selected according to their roles in mechanical construction, image processing, object detection, and system integration. The final selection of hardware specifications and software versions should be based on the implemented system.

7.1 Software Tools

7.1.1 Robot Operating System (ROS / ROS 2)

The Robot Operating System (ROS) is an open-source, flexible software middleware framework designed to simplify the construction of robust, modular robotic platforms. Rather than operating as a conventional hardware-level operating system (such as Linux or Windows) that directly schedules CPU processes and handles peripheral hardware interrupts, ROS functions as a robotic middleware or meta-operating system. It provides structured software services including hardware abstraction, low-level device control, standard implementations of core robotic algorithms, inter-process communication, package management, and an extensive suite of diagnostic and visualization tools.

Entering into this project, ROS 2 was a new technology that the project team had to learn and actively familiarise itself with. Integrating a plant-root-inspired tip-growing robot requires harmonizing disparate engineering domains: pneumatic pressure regulation, four-tendon antagonistic kinematics, high-frequency servo PWM control, deep-learning-based visual object detection, and spatial path planning. Navigating this multi-disciplinary complexity demanded a software framework that could decouple these functional modules while maintaining deterministic communication. Familiarising ourselves with ROS 2 provided the foundational architecture required to bring these independent mechanical, electronic, and algorithmic components into a cohesive, synchronized system [11], [15].

7.1.1.1 Why ROS 2? Architectural Motivation and Justification

Selecting the central software architecture was one of the critical decisions in the project. Several conventional approaches were evaluated, and the rationale for adopting ROS 2 is detailed below:

1. **Failure of Monolithic Software Architectures:** A basic approach in student prototypes is to construct a single monolithic script (e.g., in Python or C++) that sequentially captures camera frames, executes YOLO inference, calculates inverse tendon lengths, reads pressure sensors, and transmits serial servo commands. However, such a monolithic loop suffers from severe blocking vulnerabilities. If YOLO inference experiences temporary latency (e.g., dropping from 30 ms to 100 ms during a dense visual scene), the entire execution loop freezes. In a pneumatically driven tip-growing robot, even a momentary pause in tendon payout while tube eversion continues can lead to excessive tension, structural buckling, or catastrophic tube tearing. ROS 2 prevents this failure mode by executing perception, kinematics, growth management, and serial communication in isolated, concurrently running nodes managed by a multi-threaded executor.

2. **Computational Inadequacy of Standalone Microcontrollers:** Microcontrollers such as the ESP32 or Arduino are well suited for precise PWM pulse generation and low-level digital I/O. However, they lack the multi-core processing power, GPU acceleration, and memory bandwidth necessary to host deep convolutional neural networks (such as YOLO), maintain high-bandwidth video streams, and evaluate 3D coordinate transformation trees. ROS 2 enables a natural hierarchical split: resource-heavy deep learning and path planning execute on a host computer (or single-board computer), while time-critical physical actuation runs on the microcontroller, seamlessly linked via the Micro-ROS bridge.

3. **Evolution and Superiority of ROS 2 over ROS 1:** The original ROS 1 architecture depended entirely on a centralized master node (`roscore`) to coordinate node registration and message routing. If `roscore` failed, the entire robotic computational graph crashed. Furthermore, ROS 1 lacked deterministic real-time scheduling, lacked security protocols, had limited cross-platform portability, and required brittle third-party serial bridges (`rosserial`) prone to packet loss. ROS 2 resolves all of these limitations by building upon the industry-standard Data Distribution Service (DDS) protocol, eliminating the central master in favor of decentralized peer-to-peer discovery, supporting real-time deterministic execution, and introducing native microcontroller integration.

7.1.1.2 Team Familiarisation and Practical Learning Curve

Adopting ROS 2 required the project team to undergo a dedicated learning curve to transition from traditional procedural programming to distributed robotics software engineering. Key concepts mastered during the project include:

- **Workspace Architecture and Build Systems:** Learning the standard ROS 2 workspace structure (`colcon_ws/src`), structuring modular packages using `ament_python` and `ament_cmake`, and orchestrating multi-package automated builds with the `colcon` utility.

- **Custom Interface Generation:** Defining and building custom ROS 2 messages, services, and actions (e.g., custom tendon displacement arrays and eversion state status messages) that enforce strict data typing across all communicating nodes.
- **Launch Orchestration:** Developing Python-based launch files (`launch.py`) that automate the parameterized initialization of multiple nodes, sensor drivers, coordinate transform publishers, and safety watchdogs with a single command.
- **Client Libraries (`rclpy` and `rclcpp`):** Leveraging Python's client library (`rclpy`) to interface seamlessly with modern AI libraries (PyTorch, Ultralytics YOLO, and OpenCV), while retaining the capability to implement performance-critical kinematic loops in C++ (`rclcpp`).

7.1.1.3 Core Features and Communication Paradigms of ROS 2

The ROS 2 architecture provides four distinct communication paradigms that map directly onto the requirements of the growing robot system:

- **Topics (Publish–Subscribe Model):** Topics provide asynchronous, many-to-many communication channels suitable for high-throughput streaming data. In the proposed system, raw video frames captured at the growing tip are published over `/camera/image_raw` using standard `sensor_msgs/msg/Image` formats, while detected object bounding boxes and labels are published onto `/detections/bounding_boxes` using `vision_msgs/msg/Detection2DArray`.
- **Services (Request–Response Model):** Services provide synchronous, two-way client–server interactions for episodic or state-changing commands. Services are utilized in Robo-Root for operations such as zeroing the tendon servo drums, toggling the pneumatic solenoid valve state, resetting path waypoints, or requesting camera recalibration routines.
- **Actions (Goal–Feedback–Result Model):** Actions provide an asynchronous, preemptible communication mechanism specifically designed for long-duration robotic tasks. An action client dispatches a goal, the action server provides continuous periodic feedback during execution, and eventually returns a final result status. Actions are ideally suited for commanding tip-growing navigation, such as directing the robot to evert forward by 25 cm along a specified heading, with the critical capability to preempt or cancel the operation immediately if an unforeseen obstacle is detected by the vision node.
- **Parameters (Dynamic Reconfiguration):** Parameters are node-level configuration variables that can be inspected, validated, and updated dynamically at runtime without restarting the node. This enables live tuning of YOLO confidence thresholds, servo angular velocity limits, PID gains, and pneumatic chamber pressure limits.

7.1.1.4 Quality of Service (QoS) Engine

Unlike generic network protocols, ROS 2 incorporates a configurable Quality of Service (QoS) engine governed by the DDS layer. This allows the development team to tailor communication policies to the specific physical characteristics of each subsystem:

- **Reliability Policy:** The `Best Effort` policy is applied to high-bandwidth sensory streams such as the camera video feed, ensuring minimal transmission latency by prioritizing real-time frame rates over occasional packet drops. Conversely, the `Reliable` policy is enforced for critical actuation commands (servo angular targets, solenoid triggers, and emergency stop signals), guaranteeing that every command packet is acknowledged and retransmitted if dropped.
- **Durability Policy:** The `Volatile` policy is applied to live sensor streams, ensuring nodes only receive fresh data. In contrast, the `Transient Local` (latched) policy is applied to static configuration topics and coordinate frame trees, ensuring that newly initialized visualization or planning nodes immediately receive the current system state upon startup.
- **History and Depth:** Prevents message queue overflow and memory starvation on edge computing devices by bounding queue depths (e.g., `Keep Last 5`).

7.1.1.5 Dynamic Coordinate Frame Transformations (`tf2`)

A unique challenge in tip-growing robots is that their physical morphology is non-rigid and continuously expanding. As the flexible tube everts and bends under tendon tension, the spatial distance and orientation between the base station and the moving camera changes dynamically.

ROS 2 provides the `tf2` transform library to maintain coordinate frames over time in a tree structure:

$$\text{world} \longrightarrow \text{base_link} \longrightarrow \text{growing_body} \longrightarrow \text{growing_tip} \longrightarrow \text{camera_optical_frame} \quad (7.1)$$

The `tf2` library tracks time-stamped transformation matrices between each frame. When the YOLO detection node identifies an obstacle in the camera's local 2D pixel space, `tf2` enables the path planner to transform this detection into global 3D coordinates relative to `base_link`, allowing the planner to compute collision-free steering angles.

7.1.1.6 Microcontroller Interfacing and Micro-ROS

Physical actuation of the four steering servos and the pneumatic solenoid valve requires precise microsecond-level timing generated by an embedded microcontroller (ESP32). Rather than relying on custom, fragile string-based serial parsing scripts, the system employs `micro-ROS`.

Micro-ROS brings ROS 2 directly to resource-constrained microcontrollers using the Micro XRCE-DDS (eXtremely Resource Constrained Environments DDS) protocol over UART/USB. The microcontroller runs a lightweight micro-ROS client that communicates with a micro-ROS

agent hosted on the primary Linux computer. As a result, the ESP32 operates as a native participant in the ROS 2 graph, subscribing to servo command topics and publishing pressure sensor readings directly without translation intermediaries.

7.1.1.7 Development, Diagnostics, and Visualization Ecosystem

Familiarising ourselves with ROS 2 gave the project team access to a rich suite of developer tools that greatly accelerated debugging and experimental validation:

- **Introspection CLI Tools:** Utilities such as `ros2 topic hz`, `ros2 topic echo`, `ros2 node list`, and `ros2 doctor` allowed the team to monitor topic publish frequencies, inspect data payloads, and diagnose network health in real time.
- **Telemetry Recording and Replay (`rosbag2`):** High-dimensional sensor data, camera frames, and tendon commands can be logged into structured bag files and replayed deterministically offline, enabling algorithm tuning without continuous physical robot deployment.
- **3D Visualization with RViz2:** The RViz2 tool provides an intuitive graphical interface to visualize coordinate frames (`tf2`), live camera images, detected bounding boxes, and simulated tube curvatures simultaneously.
- **Simulation with Gazebo:** Facilitates digital prototyping and trajectory simulation before executing physical tests on hardware.

7.1.2 YOLO Object Detection

You Only Look Once (YOLO) is an object detection approach used to identify objects and estimate their locations within an image. The proposed system uses a pretrained YOLO model to process images captured by the camera near the growing tip.

The Ultralytics YOLO tools support object detection and inference on supported image and video sources [12], [13]. The model can produce bounding boxes, predicted class labels, and confidence scores for detected objects.

Within the proposed robot, these outputs can provide visual information to the ROS 2 navigation and steering components. The suitability of the model depends on whether its supported object classes are appropriate for the intended environment.

The specific YOLO version and pretrained model weights should be recorded once confirmed. No model accuracy or detection performance is assumed in this report.

7.1.3 OpenCV

OpenCV is a computer vision library that can support camera image acquisition and image-processing operations. Its functionality includes handling image data, resizing images, converting colour representations, and preparing frames for subsequent processing [14].

In the proposed system, OpenCV may be used to:

- Acquire or handle camera frames, depending on the camera interface.
- Convert image formats when required.
- Resize images according to the selected model's input configuration.
- Support image display and debugging during development.
- Assist in visualizing detected objects and bounding boxes.

Preprocessing should be configured to match the requirements of the selected YOLO implementation. If the inference framework already performs certain preprocessing operations, duplicating them should be avoided.

7.1.4 Programming Language

Python may be used to implement the camera-processing pipeline, invoke the pretrained YOLO model, and develop ROS 2 nodes. Its suitability depends on the selected ROS 2 distribution and the compatibility of the required software packages.

The program can receive camera frames, execute object detection, process detection results, and communicate with other ROS 2 components. If a separate microcontroller manages the servo motors or pneumatic valves, an appropriate communication interface must be implemented between the ROS 2 computer and the microcontroller.

The final report should specify the programming language and version actually used in the implementation.

7.1.5 Development and Visualization Tools

A code editor or integrated development environment may be used to develop, organize, and debug the software. ROS 2 command-line tools can assist with checking nodes, topics, and message communication. RViz2 may be used to visualize supported sensor data or other ROS 2 information during system development.

A simulator such as Gazebo may be considered if simulation is part of the project. These tools should only be listed as implemented tools if they were actually used; otherwise, they should be identified as optional development tools.

7.2 Hardware Tools

7.2.1 Pneumatic Components

The pneumatic system provides the pressure required to drive the tip-eversion mechanism. Its components may include an air pump or compressor, pressure regulator, tubing, fittings, and a suitable valve.

The pump supplies air, the regulator helps control the operating pressure, and the valve controls the air supply to the robot. The selected components must be compatible with the required pressure range, flow demand, and physical construction.

The actual component models and their pressure ratings should be documented using the corresponding manufacturer datasheets.

7.2.2 Servo Motors and Tendon Actuation

Four servo motors are proposed for actuating the four steering tendons arranged around the robot body. Coordinated tendon movement can create directional bending of the growing structure.

The servo motors must provide sufficient torque and suitable travel for the tendon mechanism. Their selection should consider the required tendon tension, cable routing, friction, and mechanical limits.

The motor model, torque rating, operating voltage, and control interface should be recorded from the datasheet for the components actually selected.

7.2.3 Camera and Sensors

A camera positioned near the growing tip provides images of the environment for object detection. The camera should provide adequate image quality and a suitable field of view for the intended operating conditions.

Additional sensors, such as a pressure sensor or inertial measurement unit (IMU), may be considered to monitor pneumatic conditions or estimate motion and orientation. Their inclusion depends on the final hardware design.

The report should distinguish sensors that are installed and tested from those that are only proposed for future development.

7.2.4 Microcontroller and Electronic Interface

A microcontroller, such as an ESP32 or another compatible controller, may be used to interface with the servo motors and pneumatic control hardware. It can receive commands from the main computer and generate the appropriate actuator control signals.

The electronic system may also include a suitable power supply, servo driver, wiring, connectors, and valve driver circuitry. Electrical ratings and interface requirements should be checked against the actual component datasheets.

Appropriate protection and control measures are necessary to prevent unintended actuator motion and excessive pneumatic pressure.

7.3 Mechanical Design Tools

Computer-aided design (CAD) software may be used to prepare the mechanical layout of the robot, including the base station, tendon-routing mechanism, camera mount, and pneumatic

connections.

The design process should consider alignment, assembly access, cable friction, structural flexibility, and the space required for the growing tube. CAD drawings can help document component placement and support fabrication.

The specific CAD package should be named in the final report according to the software actually used.

7.4 Integration of Tools

The software and hardware tools operate as interconnected parts of the overall system. The camera supplies environmental images, OpenCV can handle image-processing tasks, and the pretrained YOLO model produces object detections [12]–[14]. ROS 2 can coordinate communication between perception, navigation, and hardware-control components [11].

The hardware-control interface then communicates the required commands to the servo motors and pneumatic system. The servos actuate the steering tendons, while the pneumatic subsystem controls growth of the robot body.

The overall system therefore combines perception, decision-making, actuation, and communication. The actual level of automation depends on the implemented control logic and the availability of feedback from the robot and its sensors.

CHAPTER 8

CONCLUSION

The proposed project focuses on the design and development of a pneumatic tip-growing robot that combines flexible tube eversion, four-tendon steering, camera-based object detection, and ROS 2-based control. The objective is to establish a modular robotic system capable of controlled forward growth and directional steering, with visual perception supporting navigation decisions.

The mechanical design incorporates a flexible pneumatic tube that extends through eversion and four axial tendons arranged around the growing body. Four servo motors positioned at the base station are intended to control tendon movement and direct the growing tip in the horizontal X–Y plane. This arrangement separates the main actuation hardware from the moving tip while introducing design requirements related to tendon routing, tension, friction, and cable payout during growth.

For environmental perception, a camera mounted near the growing tip provides images to a pretrained YOLO object detection model. The model identifies objects belonging to its supported classes and produces bounding boxes, class labels, and confidence scores. These results are intended to provide visual information to the path-planning component. ROS 2 provides a modular software architecture for connecting image processing, path planning, steering control, and pneumatic growth management.

The literature review established that tip-growing robots, pneumatic eversion, steering mechanisms, sensing approaches, and simulation techniques have been investigated in previous research. These studies also highlight challenges associated with payload integration, friction, curved-path behavior, sensing reliability, and retrieval. Consequently, the proposed project focuses on integrating established robotic principles into a specific mechanical and software architecture rather than claiming a completely new growing-robot concept.

The effectiveness of the proposed system will depend on the successful integration of its mechanical, pneumatic, electronic, and software components. Future evaluation should examine growth stability, steering accuracy, tendon coordination, object detection reliability, and the response of the integrated system under different operating conditions. Simulation and physical experiments will be necessary to verify the relationship between planned trajectories and actual robot movement.

Further development may include calibrated closed-loop steering, improved obstacle-distance estimation, more reliable tendon tension management, and systematic testing of deployment and retrieval. Overall, the project provides a foundation for investigating camera-assisted navigation in a pneumatically actuated, tendon-steered growing robot. Its practical contribution will be established through implementation, experimental evaluation, and comparison against clearly defined performance criteria.

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